gpeyt1-27 · Exit gradient / boiling

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

Inside a sheet-pile cofferdam excavated in fine sand (γsat = 124 pcf, Gs = 2.66, e = 0.62), a flow net shows the last equipotential square adjacent to the excavation floor has a head loss of Δh = 2.8 ft over a flow-path length of 4.5 ft. Total head difference across the cofferdam is 16 ft and the sheet piles are embedded 14 ft below the floor.

Most nearly, the factor of safety against boiling at the exit is:

A. 3.19
B. 1.59
C. 0.63
D. 0.86

Given equations / data

> `icr = γ′/γw; iexit = Δh/l; FS = icr/iexit; γw = 62.4 pcf`

Worked solution

1. `γ′ = 124 − 62.4 = 61.6 pcf → icr = 0.987`
2. `(check: (Gs−1)/(1+e) = 1.025)`
3. `iexit = 2.8/4.5 = 0.622`
4. `FS = 0.987/0.622 = 1.59`

Final answer

1.59

Why the other values are wrong

- **3.19** — used γsat/γw for icr
- **0.63** — inverted FS
- **0.86** — used total head/embedment as the exit gradient

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.3 Flow Through Soil / §3.16.4 Flow Nets, pp. 188–190

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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